Quiz 10:-

C6:- $\int_0^{\pi/3} \sin x \ln(\cos x)\, dx$

Let $u = \ln(\cos x)$ | $dv = \sin x\, dx$

$du = \frac{1}{\cos x}\, dx$ | $v = -\cos x + C$

$\int u\,dv = uv - \int v\,du$

$\int \sin x \ln(\cos x)\, dx = \ln(\cos x)(-\cos x) - \int (-\cos x)\left(\frac{1}{\cos x}\right) dx$

$= -\ln(\cos x)\cos x + x + C$

$= -\cos x \ln(\cos x) + x \Big|_0^{\pi/3}$

$= \left[-\cos \pi/3 \ln(\cos \pi/3) + \pi/3\right] - \left[-\cos 0 \ln(\cos 0) + 0\right]$

$= \left[-\frac{1}{2}\ln(1/2) + \pi/3\right] + 0$

$= \pi/3 - \frac{\ln(1/2)}{2}$

$$\int_0^{\pi/3} \sin x \ln(\cos x)\,dx = \frac{\pi}{3} - \frac{\ln(1/2)}{2}$$

CT: $\int (3+\cos x)^2\,dx$

$= \int (9 + 6\cos x + \cos^2 x)\,dx$

$= \int 9\,dx + \int 6\cos x\,dx + \int \cos^2 x\,dx$

$= 9x + 6\sin x + \int \left(\frac{1-\sin 2x}{2}\right) dx + C$

$= 9x + 6\sin x + \int \frac{1}{2}\,dx - \int \frac{\sin 2x}{2}\,dx + C$

$= 9x + 6\sin x + \frac{1}{2}x - \frac{\cos 2x}{4} + C$

$= \frac{19}{2}x + 6\sin x - \frac{\cos 2x}{4} + C$

$\therefore \int (3+\cos x)^2\,dx = \frac{19x}{2} + 6\sin x - \frac{\cos 2x}{4} + C$

Q9:- (A)

$$\int \frac{(x-4)}{x^2-5x+6}\,dx$$

$$= \int \frac{(x-4)}{(x-2)(x-3)}\,dx$$

$$= \int \left[\frac{A}{(x-2)} + \frac{B}{(x-3)}\right] dx$$

$A(x-3) + B(x-2) = x-4$

$Ax - 3A + Bx - 2B = x - 4$

$A + B = 1$

$-3A - 2B = -4$

$\Rightarrow \quad -3A + 3B = -3$

$-3A - 2B = -4$

$B = -1$

$\therefore \boxed{A = 2}$

$x-2$	6
$3-2$	-5

$$= \int \left[\frac{2}{(x-2)} - \frac{1}{(x-3)}\right] dx$$

$$= 2\ln|x-2| - \ln|x-3| + C$$

$$\therefore \int \left(\frac{x-4}{x^2-5x+6}\right) dx = 2\ln|x-2| - \ln|x-3|$$

(B) $$\int \frac{2x+1}{x^3-4x}\,dx = \int \frac{2x+1}{x(x^2-4)}\,dx$$

$$= \int \frac{2x+1}{x(x-2)(x+2)}\,dx$$

$$= \int \left(\frac{A}{x} + \frac{B}{(x-2)} + \frac{C}{x+2}\right) dx$$

$$2x+1 = A(x^2-4) + B(x^2+2x) + C(x^2-2x)$$

$$2x+1 = Ax^2 - 4A + Bx^2 + 2xB + Cx^2 - 2xC$$

$$A + B + C = 0$$

$$2B - 2C = 2$$

$$-4A = 1 \Rightarrow \boxed{A = -1/4}$$

$$B + C = 1/4$$

$$B - C = 1$$

$$2B = 5/4 \Rightarrow \boxed{B = 5/8}$$

$$2B - 2C = 2 \Rightarrow 2(5/8) - 2C = 2$$

$$-2C = 2 - 5/4 \Rightarrow -2C = 3/4$$

$$\boxed{C = -3/8}$$

$$= \int \left[\left(-\frac{1}{4}\right)\left(\frac{1}{x}\right) + \left(\frac{5}{8}\right)\left(\frac{1}{x-2}\right) - \left(\frac{3}{8}\right)\left(\frac{1}{x+2}\right) \right] dx$$

$$= -\frac{1}{4}\ln|x| + \frac{5}{8}\ln|x-2| - \frac{3}{8}\ln|x+2| + C$$

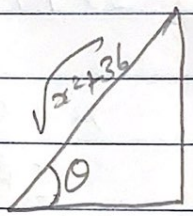
$$\therefore \int \frac{2x+1}{x^3-4x}\,dx = \frac{5}{8}\ln|x-2| - \frac{1}{4}\ln|x| - \frac{3}{8}\ln|x+2| + C$$

C8:

$$\int \frac{x^2}{\sqrt{x^2+36}}\,dx$$

$x = 6\tan\theta$

$dx = 6\sec^2\theta\,d\theta$



$$= \int \frac{36\tan^2\theta\,(6\sec^2\theta\,d\theta)}{\sqrt{36\tan^2\theta + 36}}$$

$$= \int \frac{36\tan^2\theta\,(6\sec^2\theta\,d\theta)}{6\sec\theta}$$

$$= \int 36\tan^2\theta\sec\theta\,d\theta$$

$$= 36\int \tan^2\theta\sec\theta\,d\theta$$